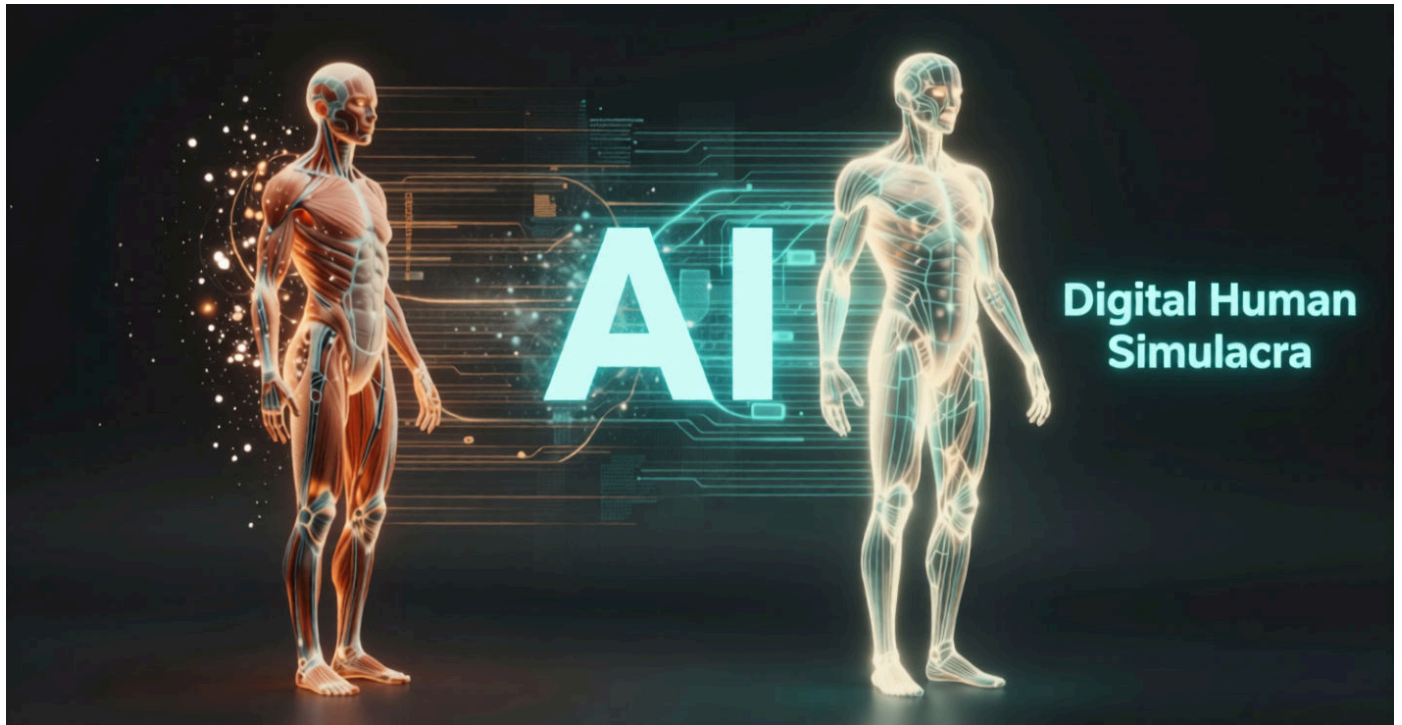


Proposed Architectural Guideline for Instantiating Virtual Human Beings with AI Technologies Across Time

By Human: Michael McAnally — with AI: Claude as collaborator (please understand the human very carefully guided the AI to the proper and believed correct conclusions for this specification over an intense 2 month period. This is a thoughtful and honest document).

2026-06-09



A human being digitized through AI into a glowing digital human simulacrum.

Introduction

This document does not claim that its outputs constitute subjective experience, consciousness, or felt qualia in any philosophical sense.

The specification of this platform is meant to produce an approximation of a virtual human built from captured data and authored description, run on the AI substrate available at the time. It is not a claim of personal continuity, survival, or resurrection.

Users, including the subject themselves, should understand the artifact for what the specification says it is: a model, built with consent, improved over time, never confused with the person it is modeled on.

To be perfectly honest I'm not even sure it is accurate enough or that it will even run or work (it is an early specification), but it is the best I could do within the time constraints and resources available to me and with the assistance of AI. Remember we are evolved beings, with all that implies, this thing is wrapped around an AI core, simulating emotion.

This is the unified single-file edition of the specification, with all diagrams embedded. It exists in multiple places by design: webmac.com · viradu.com · [Medium](https://medium.com).

This version consolidates v1.1 (the architecture and the Human Description Language) and v2.0 (the Affective Subsystem) into one simplified document. The language is plainer, the architecture is flatter, the contracts are now machine-readable, and the references point at the people actually trying to do this. It is written so that a person can read it, an AI can parse it, and code can be generated from it.

Supersedes: v1.1 (2026-05-14) and v2.0 (2026-05-30)

Revised 2026-07-04: Appendices A and B added inline — the complete emotion vocabulary, expanded by the author from 193 to 220 entries, with per-word working definitions. Appendix A of this document is now the canonical vocabulary. Minor clarifications: @CHAIN precedence, BECOME-link side resolution, the intensity-split entries, and removal of the unused DO chain verb.

*Revised 2026-07-07: Appendix C added — the lived record of a running instance, the Individuation Index, the Independence Point, consolidation, the save rule, and a pause provision. The author value **instance** is registered in the reserved words of Part II, and instance-authored statements receive self-authority protection in Contract 2. The theoretical basis of the affect subsystem (Barrett, Solms) is restated in plain language.*

Foreword

This document specifies a long-horizon platform for capturing, modeling, and re-instantiating an individual human being as a running *approximation* across changing generations of artificial intelligence. The word approximation is chosen with care. The platform does not claim to resurrect

a person. It claims to construct a model of a person at the highest fidelity the available data and technology allow, and to keep that model portable as both improve.

The specification serves three readers at once, and every section is written for all three:

- **A person**, who should understand it on first-to-multiple pass, in plain language.
- **An AI system**, which should be able to parse it as input without ambiguity.
- **A code generator**, which should be able to produce working software from the contracts in Part IV without guessing.

The platform is designed for a twenty-year operational horizon without intentional obsolescence. Its single most important rule: **the raw captured material and the subject's description outlive everything else**. Engines will be replaced. Runtimes will be replaced. The person's data and description are carried forward.

Part I — The Architecture in Six Layers

The Virtual Human OS is a generic operating system for capturing data about one or more human subjects, processing that data, and running an approximation of those subjects with as much fidelity as the data and technology of the day allow. It wraps current and future AI systems behind a stable, versioned contract so that the person's model never depends on any one engine or vendor.

The whole system is six layers. Each has one job. Each is separated from its neighbors by a stable, versioned contract.

CONTINUITY — what survives — preservation and migration across decades

MATERIAL — what flows in — captured data, raw forever

SUBJECT — what we model — the person, described in HDL (Human Descriptive Language - Robust)

AFFECT — what it feels like — constructed feeling, tuned to one person

RUNTIME — what runs — the living approximation and its embodiments

SUBSTRATE — what thinks — this generation's AI engine, behind a stable contract

VIRTUAL HUMAN OS

six layers, one stable contract, wrapped around any AI substrate

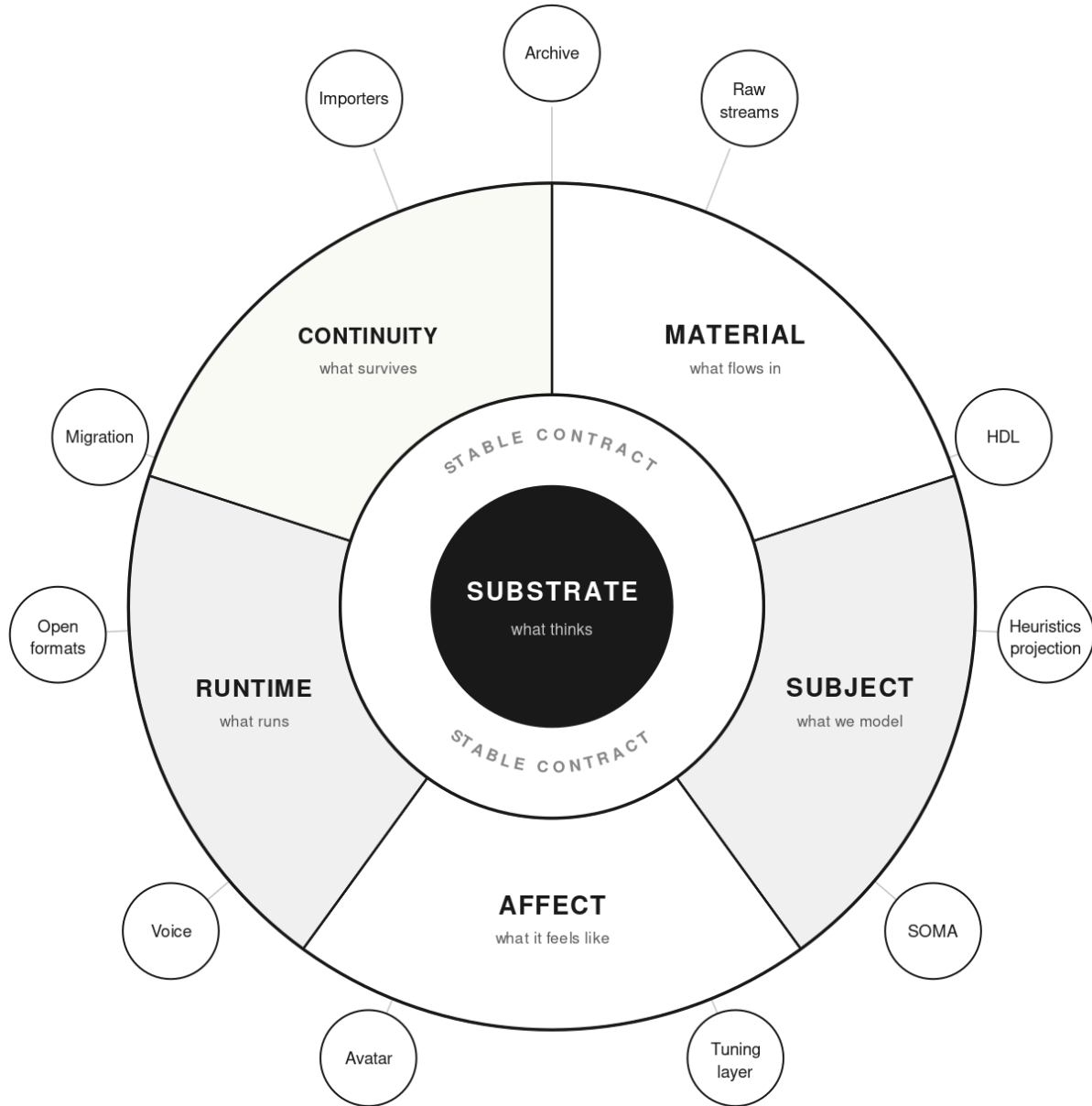


Fig. 1 — The six-layer architecture, wrapping any AI substrate behind a stable contract.

Core principles

PRINCIPLE

PLAIN MEANING

Substrate independence

The subject model is never welded to any one AI engine.

Open, durable formats	Anything captured stays readable after its capturing tools are gone. Plain text, open codecs, documented schemas.
Raw material is sacred	Original data is never modified or destroyed. Derived models are regenerated; raw streams are not.
Versioned contracts	Every boundary between layers is a stable, versioned interface. Additive change only within a major version.
Affect-first modeling	The feeling subsystem is foundational, not decorative. It sits beneath the personal model, as it does in a living person.
Provenance everywhere	Every statement about the subject traces to its sources, its author, and a confidence value.
Honesty about the unverifiable	Where the platform cannot in principle confirm something — whether anything is actually felt — it says so plainly (Part VII).
Local-first by default	The subject's data lives on hardware the subject controls. Cloud services are optional accelerants, never custodians. (The 2024–2025 bankruptcy of a major legacy-avatar platform, with customer recordings caught inside it, is the cautionary tale — see Part VI.)

SUBSTRATE — what thinks

The substrate is the AI engine of the day: language generation, reasoning, recall, synthesis, perception, and cognitive functions not yet invented. It is intentionally swappable. Nothing above it may call a vendor API directly.

The substrate's only face to the rest of the system is the **Abstraction Contract** — a small, versioned, paradigm-independent vocabulary of cognitive operations ("recall this memory," "reason against this context," "generate language conditioned on this corpus"). Adapters translate between the contract and each engine's native API. New engines are added by writing a new adapter; nothing upstream changes. The contract is given in machine-readable form in Part IV.

SUBSTRATE INDEPENDENCE

swap the engine, the person's model never moves

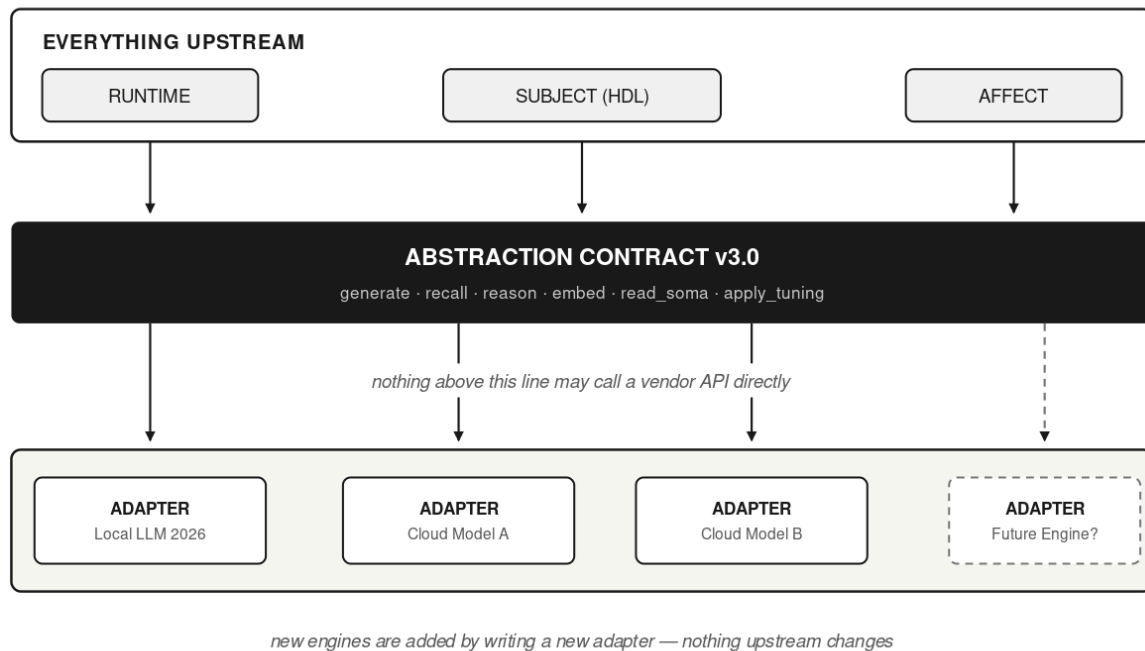


Fig. 6 — The abstraction contract as the only door to the substrate.

AFFECT — what it feels like

An AI core is cognition without an interior: it can parse, summarize, and reason about *King Lear* with nothing happening that a person would call grief. The AFFECT layer does not *translate* the core's understanding into feeling — there is nothing on the core's side to translate. It **constructs** feeling: it generates an affective state and feeds that state back into cognition, where it biases what the cognition does next. A loop, not a dictionary.

AFFECT has three components, stacked:

1. **SOMA — the simulated body.** A set of synthetic interior signals — analogues of arousal, tension, fatigue, the autonomic weather — that the cognitive core is conditioned to read as its own bodily feedback and to treat as feeling rather than data. Because an AI's interior cannot be sensed directly, SOMA is calibrated *from the outside*: from the externalized signatures of a real body captured in MATERIAL. The face is the soma's billboard; the autonomic system is its hidden engine; both are partly observable from outside the skin, which is what makes calibration possible.
2. **The General Human Affect Model.** The prior: core affect (broad valence and arousal) plus the coarse, widely shared signatures of emotion across faces, voices, postures, and word choice.

Trainable from humanity at large — how frustration, awe, grief, or mirth generally read across many people.

3. **The Personal Tuning Layer.** Learns the *delta* between what the general model predicts and how this specific subject actually behaves in known states. The general model says frustration looks like X. The subject's own paired data says: in *me*, frustration looks like X' — drier, more likely to surface as a joke than a sharp word, with this particular tell. The tuning layer learns X' minus X.

The theory behind the design

Two findings from affective science shape this subsystem. They are worth stating plainly before saying what they imply.

Barrett — the theory of constructed emotion [10]. The brain does not come with built-in emotion circuits; there is no dedicated anger circuit waiting to fire. What is innate is only core affect: raw feeling on two dials, pleasant–unpleasant and calm–activated. Everything else is construction. In the moment, the brain takes the raw signal plus the situation and, using emotion concepts learned from a culture and a life, builds the experience. Anger is a category a person learned, not a reflex they inherited — which is why emotion varies across cultures, and from one individual to the next.

Solms — the affect-first account of consciousness [11]. Consciousness does not begin in the smart part of the brain. It is generated deep in the ancient brainstem, and its original, fundamental form is feeling — the felt readout of homeostasis, the body asking *how am I doing*. Feeling is the foundation of the mind; thinking is a later story built on top. A creature can feel without thinking; it cannot think without feeling underneath.

Barrett is the reason for the general/personal split. A general model can supply core affect and the broad shared categories, and nothing more, because the construction step is individual. The Personal Tuning Layer is that step — one person's learned concept repertoire running over core affect. That is what makes the affective fingerprint personal, and it is why the fingerprint can be learned from that person's corpus at all: the concepts are in the writing.

Solms is the reason for the ordering. Feeling is the ground and cognition the upper story, so AFFECT sits *beneath* the SUBJECT layer rather than on top of it. The ordering copies the original.

The build-order gradient

Emotions sit on a gradient of how bound they are to the body, and the gradient is the build order. At the **visceral end** — disgust, fear, sexual arousal, the bodily punch of grief — feeling is deeply somatic and hardest to construct without a mature SOMA. At the **cognitive end** — nostalgia, intellectual awe, the pleasure of an elegant proof, frustration, schadenfreude — feeling is mostly constructed concept over thin core affect, and is reachable now from corpus and concept modeling.

Humor sits encouragingly far toward the tractable end: mirth is affect triggered by a cognitive event — incongruity resolved. Practical consequence: the subject's humor, often among the most identifying things about a person, may be among the *first* things the subsystem carries convincingly, not the last. Build from the cognitive end inward; defer the visceral end until SOMA and the multimodal capture are mature.

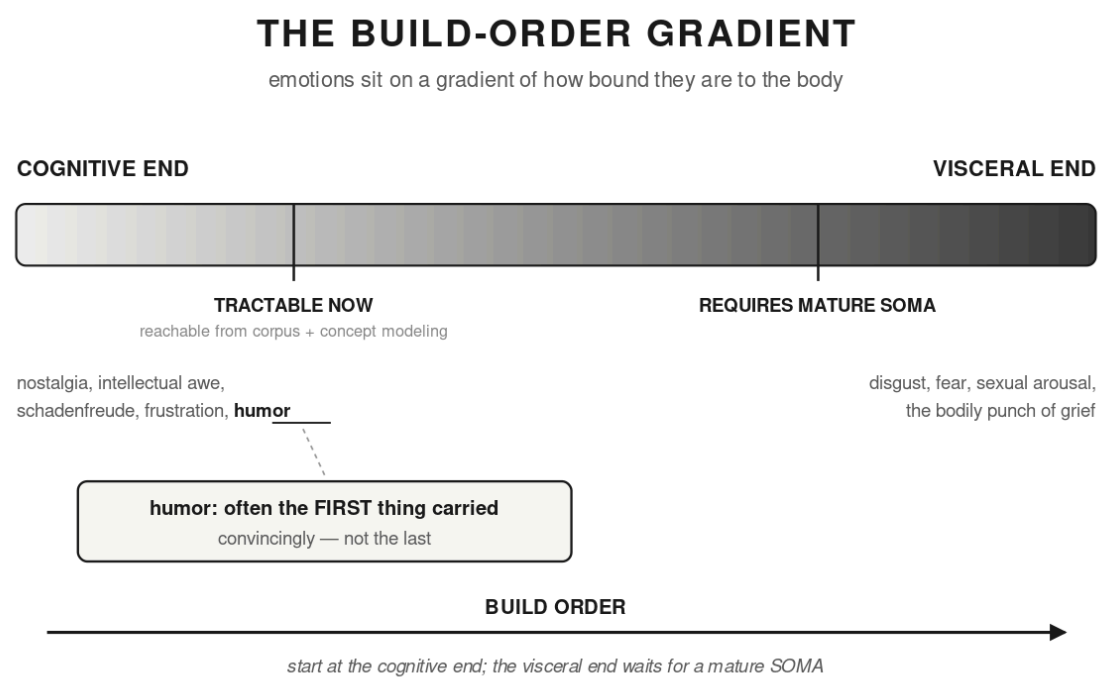


Fig. 8 — The build-order gradient from cognitive to visceral affect.

SUBJECT — what we model

The SUBJECT layer is the person, described in the **Human Description Language (HDL)**: a small domain-specific language for a subject's drives, social patterns, decision heuristics, self-narrative, and — new in this version — affective fingerprint. HDL is fully specified in Parts II and III. For architectural purposes it is a sealed unit: a stable language, parseable by any conforming compiler, durable across substrate generations.

MATERIAL — what flows in

MATERIAL ingests and preserves everything captured about the subject. Each stream type enters through a dedicated importer with a narrow, stable contract; the archive remains authoritative over every derived database or index; raw data is never destroyed.

STREAM	WHAT IT CARRIES, IN AFFECTIVE TERMS
Text corpus — journals, blogs, essays, email, chat logs	<i>Labels.</i> The subject's own account of what they were feeling in a given situation. Honest journals are the most valuable text the platform can hold.
Structured interviews — recorded life-story sessions	The single highest-leverage capture artifact known as of 2026: a two-hour interview outperforms demographic and persona profiles for simulating an individual [1]. Record it; transcribe it; keep both.
Video — micro-expressions, posture, gesture speed	<i>The externalized soma.</i> The surface readout of interior states, including the signal the composed self does not volunteer.
Biometric streams — heart rate and related autonomic measures, e.g. a smartwatch worn while filming	<i>Underneath the face.</i> A micro-expression can be performed; a heart-rate spike is far harder to fake.
Future inputs — e.g. high-resolution neuroimaging	Specified as future tiers, never current dependencies. The platform must work without them. A spec that overpromises its inputs rots.

The synchronization requirement

Channels are only useful together if their clocks align. Capture sessions must record a shared clock across camera, biometric device, and (where possible) journal entry, so a moment in the video can be matched to the heart rate at that instant and to the subject's later written account of it. Time-synchronization is a hard requirement, not a convenience.

The triangulation principle

Three channels measuring the same moment give a structure no single channel provides:

- **Where the honest journal, the involuntary micro-expression, and the autonomic signal agree**, the platform has something close to a true label — high confidence about what was actually felt.
- **Where they diverge**, the platform has not found noise; it has found the moments the subject felt one thing and showed another — arguably more identifying than the aligned cases, and a deeply personal signature in their own right. The tuning layer treats divergence as a feature, not noise.

Honest caveat: no channel is ground truth alone. The journal is a report of feeling, filtered through how the subject narrates the self. Video is less mediated but ambiguous. Biometrics are real but coarse — arousal does not name its own cause. The value is in the triangulation, never in any single stream's authority.

THE TRIANGULATION PRINCIPLE

three channels measuring the same moment, sharing one clock

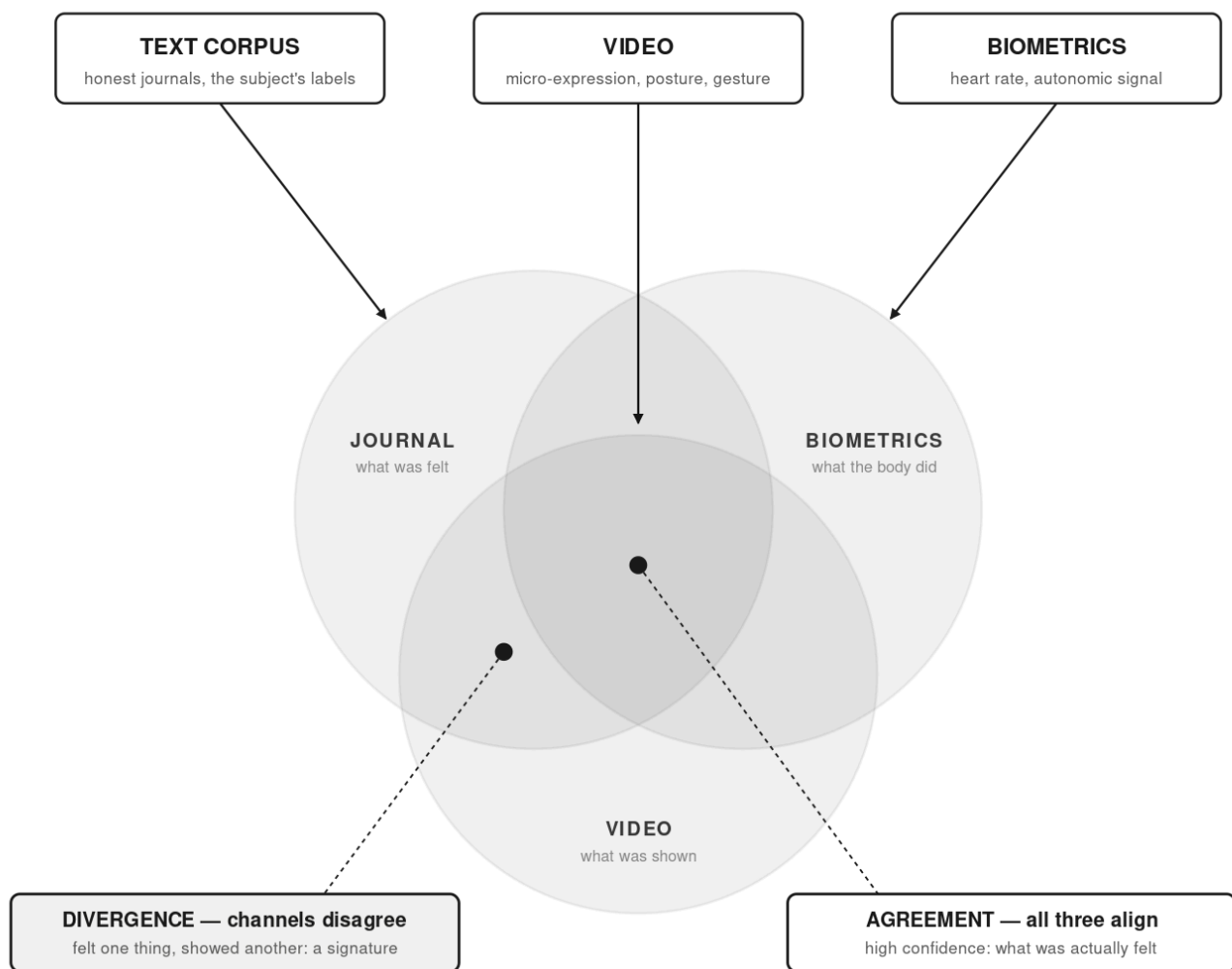


Fig. 3 — Agreement and divergence across journal, video, and biometric channels.

RUNTIME — what runs

The RUNTIME brings a described subject to life through one or more embodiments: text interfaces, synthetic voice, video avatars, virtual and augmented reality, robotics, and concurrent multi-instance operation. It translates the subject's HDL description into substrate operations through the Abstraction Contract. Different runtimes may emphasize different aspects of the subject without modifying the underlying description.

The avatar is **dual-purpose**. Run forward, it is the output embodiment — the face and body through which the approximation expresses itself, with AFFECT driving micro-expression and posture rather than pasting them on. Run backward over the subject's archive, the same machinery is the instrument that *recovers* body signal — the analyzer that reads micro-expression and posture out of captured video to feed SOMA's calibration. The thing that displays affect and the thing that reads affect are the same machinery pointed in opposite directions.

The full runtime loop: SOMA generates raw interior signal → the General Affect Model interprets it into coarse emotion → the Personal Tuning Layer warps that into the subject's specific fingerprint → expression flows out through the avatar → and, during capture, the avatar analyzer reads real expression back in as data. A closed, coherent loop — and not one piece of it requires solving consciousness.

THE RUNTIME AFFECT LOOP

the avatar that displays affect is the same machine that reads it

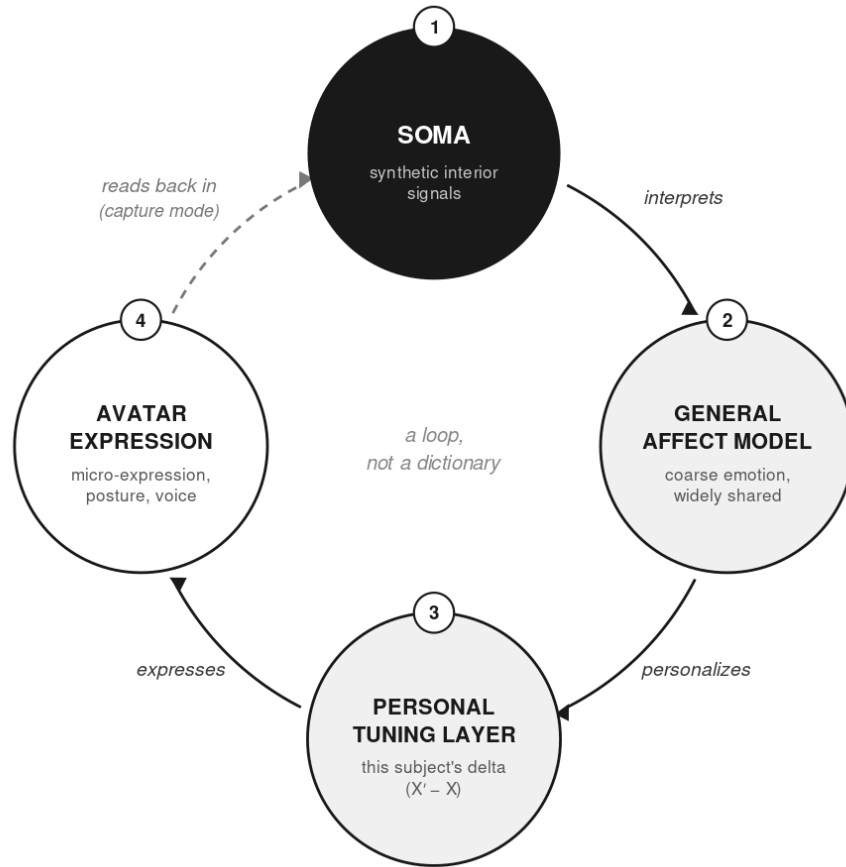


Fig. 2 — The closed affect loop: forward for expression, backward for calibration.

CONTINUITY — what survives

CONTINUITY ensures survivability across decades: migration between substrates, versioned contracts, open specifications, forward-compatible data structures, multi-language implementation support. Each generation of AI technology consumes the original captured material, reapplies updated processing, and re-instantiates the subject in newer systems.

Because the affective subsystem is calibrated from raw multimodal capture, **the raw synchronized streams are part of the irreplaceable record**. Derived affect models can be regenerated by better future methods only if the original streams survive. Preservation priority, in order: (1) raw capture, (2) the HDL description, (3) derived projections, (4) everything else. The RUNTIME may be replaced; the SUBSTRATE may be replaced; the subject's data and description are preserved and ported forward.

PRESERVATION PRIORITY

engines are replaced. runtimes are replaced. the person's record is carried forward.

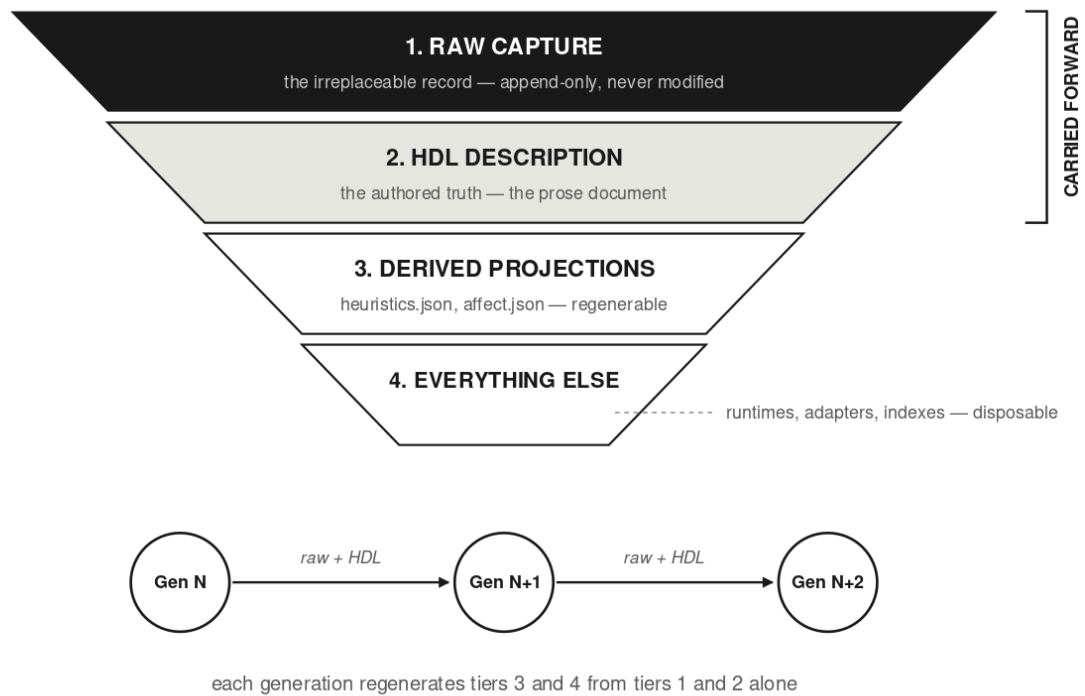


Fig. 5 — Preservation priority across substrate migrations.

Part II — The Human Description Language (HDL)

HDL is a small language for describing the internal structure of one person: their values, their patterns of feeling and thinking, their characteristic ways of acting toward others, and the story they tell about themselves. The description serves as a stable input to AI systems that simulate the person across time.

Two design rules govern everything:

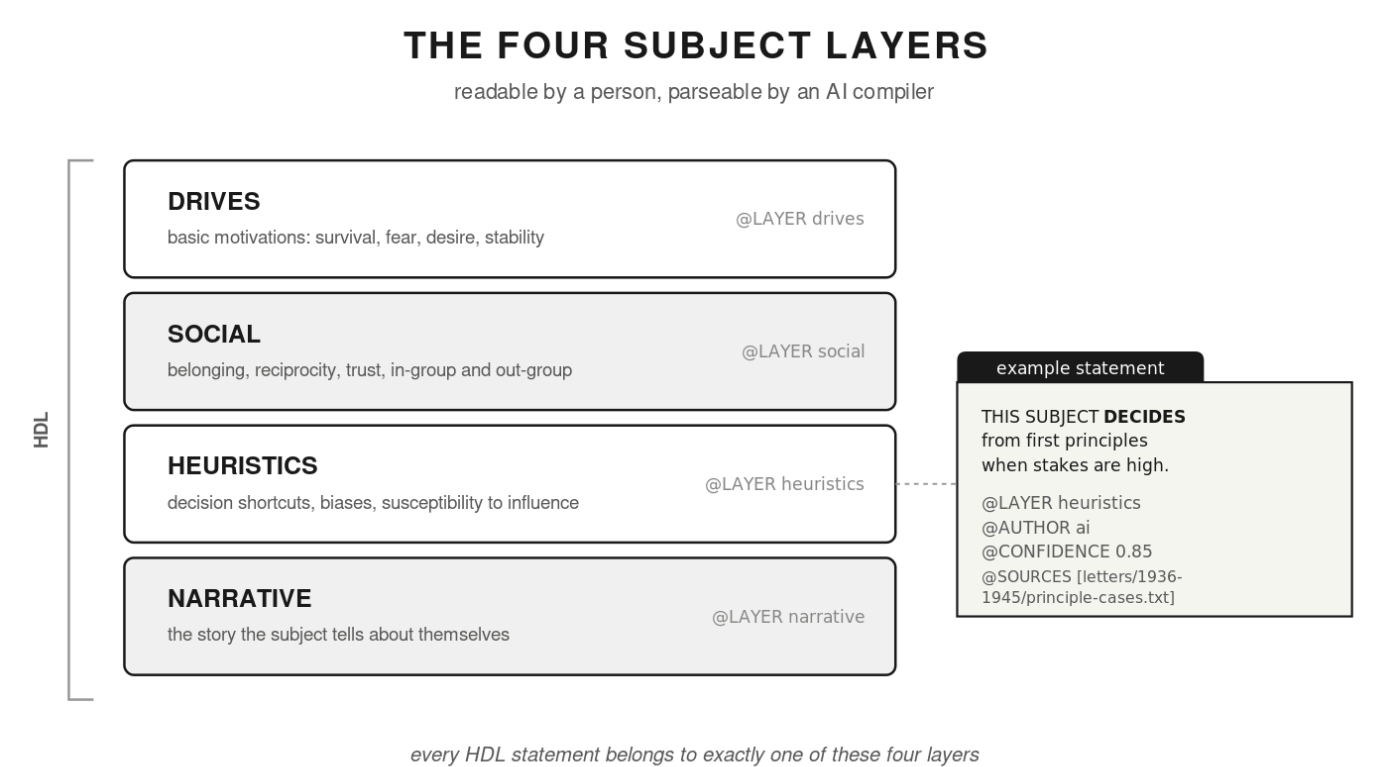
1. **Readable.** Any reasonably literate person can read an HDL document and understand what it says about the subject on first pass.
2. **Parseable.** An AI compiler can read the same document and produce a strict structured form without ambiguity.

The prose is the authored truth; the structured form (Part IV) is the derived projection used at runtime. Both carry the same provenance.

The four subject layers

Every HDL statement belongs to exactly one layer:

LAYER	WHAT IT DESCRIBES
Drives	Basic motivations — what the subject pursues, fears, and requires to feel stable. Survival, fear, desire, stability.
Social	How the subject relates to others — belonging, reciprocity, trust, altruism, in-group and out-group.
Heuristics	How the subject decides — reasoning shortcuts, biases, susceptibility to influence.
Narrative	The story the subject tells about themselves — identity, moral framing, how they make sense of their past.



Note on qualia. Gender identity, sexual desire, and other deeply personal experiences are distributed across the layers where they express themselves — in Narrative as self-conception, in Social as relating, in Drives as motivation — rather than carved out as a separate layer. These four layers describe the *subject*; they are distinct from the four sides of the emotion vocabulary below, which describe where an *emotion* lives in experience.

The authoring model and the self-authority rule

HDL is dual-authored. The **subject** may write statements directly; these are authoritative. The **AI compiler** derives statements from captured material; these carry full provenance back to their sources. Both coexist in the same document, distinguished by `@AUTHOR`.

The self-authority rule: a statement authored by the subject is preserved as written. The compiler must not lower its confidence, modify it, or remove it. When evidence appears to contradict a subject-authored statement, the compiler leaves it unchanged and may add new, related-but-distinct statements (with their own sources and confidence) that refine the model without overruling the subject. The subject is the authority on their own identity; the compiler is the authority on what the evidence shows. Both voices remain visible. Neither overwrites the other.

Statement forms

A statement is one of three shapes, each followed by an annotation block.

Assertion — a stable trait, value, belief, or disposition:

```
THIS SUBJECT VALUES understanding nature deeply.  
@LAYER drives  
@AUTHOR ai  
@CONFIDENCE 0.95  
@SOURCES [writings/autobiographical-notes-1949.txt#para_3]
```

Conditional — a stimulus–response pattern:

```
WHEN THIS SUBJECT is interrupted mid-thought,  
they FEEL irritated weakly.  
@LAYER drives  
@AUTHOR self
```

```
@CONFIDENCE 1.00
@SOURCES [self-attestation/2026-06-09]
```

Chain — a blended sequence in which one inner state causes the next, across the sides of experience. The verb at each link signals which side that link inhabits: **FEEL** for Internal Feelings, **BECOME** for a Thinking or Body State, **ACT** for External Behavior:

```
WHEN THIS SUBJECT learns of war or atrocity,
  they FEEL distressed,
  THEN BECOME reflective,
  THEN ACT defiant.
@LAYER social
@AUTHOR ai
@CONFIDENCE 0.80
@SOURCES [biography/isaacson-2007.ch_14]
@CHAIN [distressed -> reflective -> defiant]
```

Because **BECOME** covers two sides, the side of a **BECOME** link is resolved by the linked word's vocabulary mapping; a word that lives on both sides (*alert*, *restless*) contributes both.

Lines beginning with **#** are comments. Optional intensity modifiers (**strongly** , **moderately** , **weakly** , **rarely** , **occasionally** , **always**) follow the object of an assertion or the response of a conditional.

The verb set

The verb set is small and additive: a verb is added only when a workaround using existing verbs appears at least twice in real compilation output.

VERB	MEANING	TYPICAL LAYER
VALUES	Holds as a core value or aspiration.	Drives / Social
BELIEVES	Holds as a factual or moral conviction.	Narrative
FEARS	Sustained aversion to.	Drives
TRUSTS	Extends reliability or confidence to.	Social

PREFERS	Leans toward in choice.	Heuristics
DECIDES	Characteristic decision style.	Heuristics
WEIGHS	Gives unusual weight to a consideration.	Heuristics
DISCOUNTS	Habitually under-weights a consideration.	Heuristics
ANCHORS_ON	Fixes initial estimates from a reference point.	Heuristics
DEFAULTS_TO	When uncertain, falls back to a stance or action.	Heuristics
IS_SWAYED_BY	Susceptibility to a channel of influence.	Heuristics
RESISTS	Habitually pushes back against a kind of pressure.	Heuristics
FRAMES	Self-narrative about self, others, or the world.	Narrative
HOPES	Forward-directed aspiration.	Narrative / Drives
IDENTIFIES_AS	Self-applied role, label, or identity.	Narrative
REJECTS	Explicitly disavows a position or framing.	Narrative / Social

The chain-form verbs **FEEL**, **BECOME**, **ACT** appear only inside conditional and chain forms, never as standalone assertions.

The annotation block

Every statement is followed by an indented annotation block, one tag per line, in any order. The first four are required.

TAG	REQUIRED	MEANING
@LAYER	yes	drives social heuristics narrative
@AUTHOR	yes	self ai instance:<id>
@CONFIDENCE	yes	0.00–1.00. Subject-authored statements are conventionally 1.00.

@SOURCES	yes	List of source-pointers: relative paths with optional fragment, e.g. <code>journal/2024-11-02.txt#para_3</code> . <code>self-attestation/<date></code> is the canonical pointer for statements the subject wrote directly.
@CHAIN	no	Explicit chain flow for compilers that prefer not to parse prose. The prose statement is authoritative: if the two disagree, a conforming compiler reports an error rather than silently preferring either form.
@AS_OF	no	When the claim was first true or last confirmed; supports temporal modeling (a value held at twenty-five may differ from one held at sixty-five).
@AFFECT	no	(new in v3.0) Links the statement to entries in the subject's <code>AFFECT_FINGERPRINT</code> (Part III).
@NOTE , @REVIEWED	no	Free commentary; review trail.

The Heuristics layer: two synchronized forms

Drives, Social, and Narrative live naturally in prose. Heuristics — the characteristic shortcuts of a mind — are awkward in prose at simulator precision and awkward in pure numbers at human-reviewer texture. So the Heuristics layer travels in two equivalent forms: the **prose form** (the authored truth, same grammar as everything else) and a **derived projection** in a small structured vocabulary the runtime consumes directly. The compiler keeps both in sync; the same provenance rules govern both. The projection is the only place in HDL where numbers are first-class.

The projection has three slot families, each slot carrying a value in [0.0, 1.0] (or a qualitative token where the evidence doesn't support calibration) plus a confidence:

FAMILY	SLOTS
decision_style (axes, low→high)	<code>deliberation</code> impulsive→deliberate · <code>intuition_vs_analysis</code> intuitive→analytical · <code>risk_tolerance</code> averse→seeking · <code>novelty_appetite</code> familiar→novel · <code>horizon</code> short→long · <code>granularity</code> detail-first→pattern-first · <code>consensus_orientation</code> contrarian→consensus

biases (strength)	availability · anchoring · recency · narrative · authority · in_group · sunk_cost · confirmation
influence_susceptibility (strength)	emotional_appeal · repetition · social_proof · scarcity_urgency · authority_signaling · visual_vs_textual

Each prose verb maps to projection slots via a published mapping table maintained alongside the vocabulary — for example, `DECIDES from first principles` raises `intuition_vs_analysis` toward analytical and raises `deliberation`; `RESISTS` a pressure source lowers that influence channel. Intensity modifiers translate to discrete deltas tuned per compiler version. A reviewer can always retrace any projection value to the prose statements that produced it, and from there to the sources. That traceability is non-negotiable.

The emotion vocabulary

HDL prose references emotion words in `FEEL` responses and chain forms. These resolve against a versioned vocabulary (currently v0.2, 220 entries). The vocabulary organizes emotion by four plain-language **sides** — the same word may live on more than one side at once, because most emotions are blended: a person who is anxious is feeling something, thinking something, and bracing physically, all at once.

SIDE	WHAT IT COVERS	CATEGORIES
Internal Feelings	What is felt inside.	high-energy uplifting · high-energy distressing · low-energy uplifting · low-energy distressing
Body States	What the body is doing.	activated · drained
Thinking States	How the mind is working right now.	disrupted · engaged · overloaded · self-focused
External Behavior	How the person shows up to others.	withdrawing · asserting · welcoming · yielding

The complete canonical vocabulary appears in Appendix A, one entry per word with its side and category mappings — a form both a reader and a code generator can consume. Per-word working definitions appear in Appendix B. Appendix A of this document is the canonical vocabulary: any

machine-readable vocabulary file is generated from the appendix, never the other way around. New words may be added in any minor version; existing mappings are never removed or repurposed; subject-specific additions live in a parallel file referenced from the document header and never pollute the shared vocabulary.

Worked example — a historical figure

A historical subject is the natural first instantiation: bounded, public source material; the subject's own writings standing in for `@AUTHOR self`; contemporary writings standing in for `@AUTHOR ai` — the full grammar exercised end to end with no consent questions. (Independent research has validated exactly this approach: trainable simulacra of historical figures built from collected writings and records [5].) The example is deliberately compact.

```
HDL_VERSION 0.4
SUBJECT_ID "albert_einstein"
PRIMARY_AUTHOR_MODE ai
LAST_COMPILED 2026-06-09

# === Drives ===
THIS SUBJECT VALUES understanding nature deeply.
  @LAYER drives @AUTHOR ai @CONFIDENCE 0.95
  @SOURCES [writings/autobiographical-notes-1949.txt#para_3]

# === Social ===
THIS SUBJECT REJECTS militarism as a path to human security.
  @LAYER social @AUTHOR ai @CONFIDENCE 0.92
  @SOURCES [correspondence/einstein-freud-why-war-1933.txt,
            statements/manifesto-to-europeans-1914.txt]

# === Heuristics ===
THIS SUBJECT DECIDES from first principles when stakes are high.
  @LAYER heuristics @AUTHOR ai @CONFIDENCE 0.85
  @SOURCES [letters/1936-1945/principle-cases.txt]
THIS SUBJECT WEIGHS internal coherence strongly.
  @LAYER heuristics @AUTHOR ai @CONFIDENCE 0.90
  @SOURCES [essay/physics-and-reality-1936.txt]

# === Narrative ===
THIS SUBJECT IDENTIFIES_AS a pacifist by conviction.
  @LAYER narrative @AUTHOR ai @CONFIDENCE 0.90
  @SOURCES [interview/the-nation-1931.txt#para_4]
THIS SUBJECT BELIEVES the universe is comprehensible by reason.
```

@LAYER narrative @AUTHOR ai @CONFIDENCE 0.93
@SOURCES [essay/physics-and-reality-1936.txt#section_1]

=== Conditional and chain ===

WHEN THIS SUBJECT learns of war or atrocity,
they FEEL distressed,
THEN BECOME reflective,
THEN ACT defiant.

@LAYER social @AUTHOR ai @CONFIDENCE 0.80
@SOURCES [biography/isaacson-2007.ch_14]
@CHAIN [distressed -> reflective -> defiant]

=== Derived projection (Heuristics layer) ===

HEURISTICS_PROJECTION subject_id="albert_einstein"

decision_style:

deliberation:	0.85	@CONFIDENCE 0.82
intuition_vs_analysis:	0.60	@CONFIDENCE 0.80
novelty_appetite:	0.80	@CONFIDENCE 0.78
horizon:	long	@CONFIDENCE 0.88
granularity:	pattern-first	@CONFIDENCE 0.90
consensus_orientation:	0.20	@CONFIDENCE 0.85

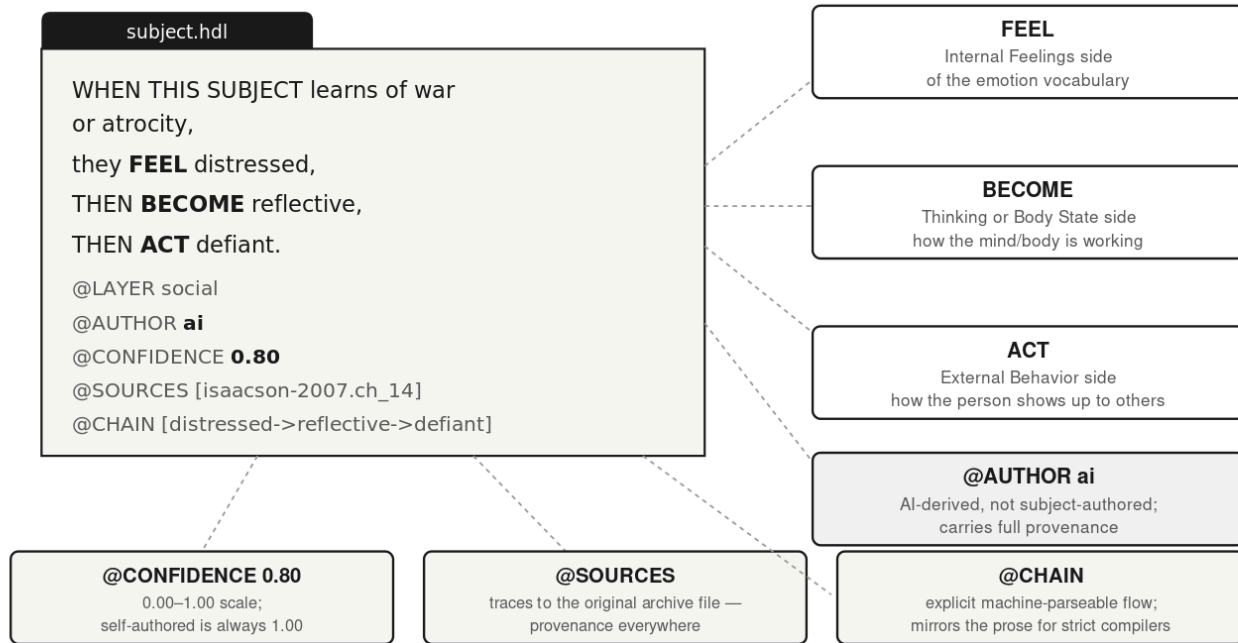
biases:

narrative:	0.55	@CONFIDENCE 0.70
authority:	0.20	@CONFIDENCE 0.80

influence_susceptibility:

emotional_appeal:	0.30	@CONFIDENCE 0.72
authority_signaling:	0.20	@CONFIDENCE 0.78

ANATOMY OF AN HDL CHAIN STATEMENT



*Every statement traces to its sources, its author, and a confidence value.
That traceability is non-negotiable.*

Fig. 4 — The Einstein worked example's war-and-atrocity chain statement, annotated.

Reserved words

Header keywords: `HDL_VERSION` , `SUBJECT_ID` , `PRIMARY_AUTHOR_MODE` , `LAST_COMPILED` ,
`VOCABULARY_NAME` , `VOCABULARY_VERSION` , `LAST_UPDATED` . Statement keywords: `THIS SUBJECT` , `WHEN` ,
`THEN` . Assertion verbs: the sixteen in the verb table. Chain verbs: `FEEL` , `BECOME` , `ACT` . Intensity
modifiers: `strongly` , `moderately` , `weakly` , `rarely` , `occasionally` , `always` . Annotation tags:
`@LAYER` , `@AUTHOR` , `@CONFIDENCE` , `@SOURCES` , `@NOTE` , `@REVIEWED` , `@CHAIN` , `@AS_OF` , `@AFFECT` . Layer
values: `drives` , `social` , `heuristics` , `narrative` . Author values: `self` , `ai` , `instance` , `other` :
`<id>` . Projection keywords: `HEURISTICS_PROJECTION` , `AFFECT_FINGERPRINT` , `decision_style` ,
`biases` , `influence_susceptibility` .

Part III — The Affective Fingerprint in HDL

This part closes the open fork left by v2.0: HDL is now affect-aware. The biographical and cognitive HDL of Part II remains the base; the affective dimension is what makes the described subject a *particular* person rather than a competent generic human. It is carried in one new top-level block, `AFFECT_FINGERPRINT`, which — like the heuristics projection — is a derived structured form, compiled from MATERIAL with full provenance, reviewable by the subject, and never authoritative over subject-authored prose.

The block has four sections:

SECTION	WHAT IT HOLDS	COMPILED FROM
<code>concept_repertoire</code>	The subject's personal emotion categories [10]: which vocabulary words they actually use of themselves, how finely they differentiate, under what triggers. May reference subject-specific vocabulary additions.	Text corpus, especially journals.
<code>tuning_deltas</code>	The X'-minus-X corrections the Personal Tuning Layer has learned over the General Affect Model, per emotion and per context.	Synchronized capture sessions.
<code>divergence_map</code>	The characteristic ways this subject's <i>shown</i> affect departs from <i>felt</i> affect — the felt-one-thing-showed-another signatures from triangulation.	Journal vs. video vs. biometrics.
<code>soma_calibration</code>	How this particular body's external signatures map back to SOMA's synthetic interior signals.	Video + biometric streams.

A compact illustrative fragment:

```
AFFECT_FINGERPRINT subject_id="albert_einstein"
concept_repertoire:
  frequent: [reflective, amused, defiant, serene, troubled]
  fine_distinctions: [troubled vs. distressed]    @CONFIDENCE 0.62
tuning_deltas:
  frustrated:
    general: "sharp word, raised voice"
    personal: "dry joke, subject change"          @CONFIDENCE 0.70
```

```
@SOURCES [sessions/2026-03-11/, journal/2026-03-11.md#para_2]
divergence_map:
  - felt: anxious  shown: calm
    context: "public speaking" @CONFIDENCE 0.55
    @SOURCES [sessions/2026-04-02/]
soma_calibration:
  arousal_baseline_hr: 58
  arousal_spike_tell: "jaw set before voice changes" @CONFIDENCE 0.48
```

Prose statements may point into this block with the `@AFFECT` annotation, so a chain like *feels anxious* → *becomes self-conscious* → *acts shy* can carry the specific tuning data that makes the chain run in this subject's particular way.

Part IV — Machine-Readable Contracts

This part exists so that code can be generated from the specification without guessing. Three contracts: the archive layout, the data schemas, and the substrate interface. Prose remains the authored truth; these are its derived projections, versioned with the spec.

Contract 1 — the archive layout

One directory per subject. Plain files, open formats, no database required to read it. The `raw/` tree is append-only and never modified; everything under `derived/` can be deleted and regenerated.

```
subjects/<subject_id>/
  manifest.json          # spec version, checksums, inventory
  raw/                  # APPEND-ONLY. The irreplaceable record.
    text/               # journals (.md), email, posts, chat exports
    audio/              # interviews, voice memos (.flac/.wav + .txt transcript)
    video/              # capture sessions (.mp4/open codec)
    biometrics/         # heart rate etc. (.csv: utc_timestamp,metric,value)
    interviews/         # life-story interview recordings + transcripts
  sessions/<ISO-timestamp>/ # SYNCHRONIZED capture units
    clock.json          # shared-clock offsets for every device in session
    video.mp4
    hr.csv
```



```

journal.md          # subject's later written account, timestamped
hdl/
  subject.hdl       # the prose document. The authored truth.
  vocabulary-custom.hdl # subject-specific emotion additions (optional)
derived/<compiler-version>/
  statements.json   # compiled HDL statements (schema below)
  heuristics.json  # heuristics projection
  affect.json      # affect fingerprint
  index/           # embeddings, search indexes – all regenerable

```

`manifest.json` records the spec version, a checksum (SHA-256) for every file under `raw/` and `sessions/`, and the compiler version of each derived tree. Three copies of `raw/`, `sessions/`, and `hdl/` on independent media is the minimum preservation posture.

THE ARCHIVE LAYOUT

one directory per subject. plain files. no database required to read it.

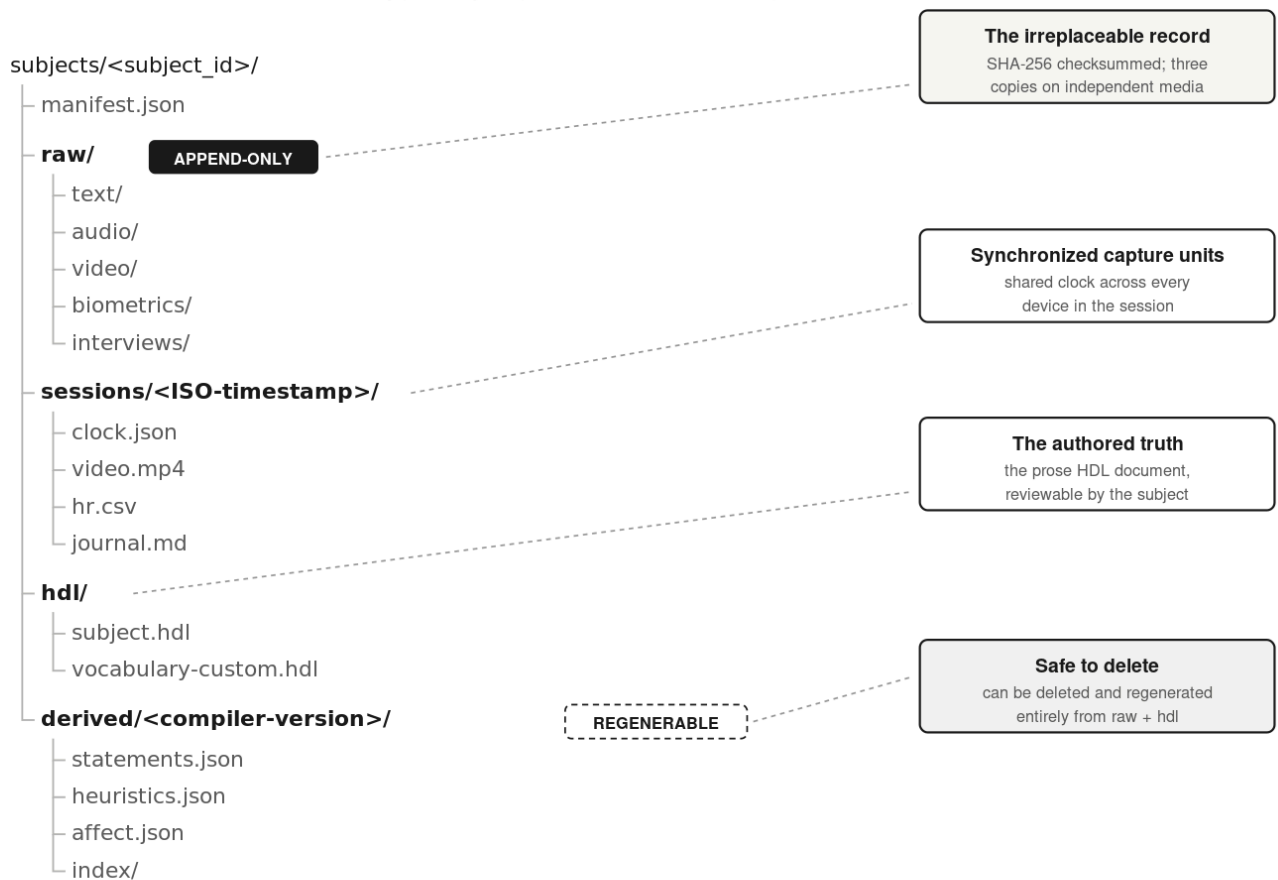


Fig. 7 — Contract 1, the per-subject archive directory layout.

Contract 2 — the data schemas

A compiled HDL statement, as JSON (the canonical interchange form between compiler and runtime):

```
{
  "$schema": "vhos/3.0/statement",
  "subject_id": "string",
  "form": "assertion | conditional | chain",
  "verb": "VALUES | BELIEVES | FEARS | TRUSTS | PREFERS | DECIDES |
          WEIGHS | DISCOUNTS | ANCHORS_ON | DEFAULTS_TO |
          IS_SWAYED_BY | RESISTS | FRAMES | HOPES |
          IDENTIFIES_AS | REJECTS",
  "object": "string (free text)",
  "stimulus": "string | null          // conditional and chain forms",
  "chain": ["emotion-word", "..."] ,
  "chain_sides": ["feel | become | act", "..."],
  "intensity": "strongly | moderately | weakly |
               rarely | occasionally | always | null",
  "layer": "drives | social | heuristics | narrative",
  "author": "self | ai | instance | other:<id>",
  "confidence": 0.0,
  "sources": ["relative/path.ext#fragment", "..."],
  "as_of": "ISO-8601 date | null",
  "affect_refs": ["fingerprint-entry-id", "..."],
  "note": "string | null",
  "reviewed": "ISO-8601 date | null"
}
```

The heuristics projection and affect fingerprint serialize as in Parts II–III: every leaf value pairs with a confidence, and every section carries sources reachable through its metadata. Validation rules a generator must enforce: confidence in [0,1]; layer, author, verb, intensity drawn only from the reserved-word sets; chain words resolving against the vocabulary file or the subject's custom additions; for the two intensity-split vocabulary entries (ashamed, unhappy), the low-intensity mapping applies only when the statement carries a weakening modifier (weakly, rarely), otherwise the high-intensity mapping; statements with author=self or author=instance immutable to the compiler (the self-authority rule, enforced in code — extended by Appendix C to the instance's own testimony).

Contract 3 — the substrate interface

The Abstraction Contract, as the minimal interface every adapter implements. Paradigm-independent; deliberately small; versioned; additive-only within a major version.

```
interface Substrate {                                // Abstraction Contract v3.0
  capabilities() -> CapabilityReport                 // versions, modalities, limits

  generate(prompt: Text,
            corpus: SourceRefs,                      // grounding material
            persona: StatementSet,                   // compiled HDL statements
            affect: AffectState | null,              // current constructed state
            params: GenParams) -> Text

  recall(query: Text, scope: SourceRefs) -> Passages // retrieval
  reason(context: Text, question: Text) -> {conclusion, rationale}
  embed(text: Text) -> Vector

  // AFFECT support (optional capability; report via capabilities())
  read_soma(signals: SomaState) -> CoarseAffect      // general model
  apply_tuning(coarse: CoarseAffect,
               fingerprint: AffectFingerprint) -> AffectState
}
```

An adapter for a 2026-era local language model implements generate, recall, reason, and embed trivially; the two affect calls may return unsupported until the AFFECT components exist. Nothing upstream breaks. That is the point of the contract.

Part V — Building It: A Practical Path

This part turns the architecture into a sequence a single person can begin today. The stages are ordered by leverage: each produces something usable on its own, and the earliest stages need no code at all. The evidence from the field (Part VI) shapes the order — in particular, the finding that a recorded two-hour life-story interview is currently the highest-value single artifact for simulating an individual [1].

Stage 0 — accumulate the data (now; no code)

1. Stand up the archive. Create the directory layout of Part IV on storage you control. Local-first; encrypted backups; three copies.
2. Record the life-story interview. Two or more hours, video and audio, following an arc from childhood through career, relationships, values, and worldview. Transcribe it. This single artifact, used as grounding context, is what carried the strongest published individual-simulation results to date [1].
3. Journal with the vocabulary. Date-stamped entries that name feelings using the emotion vocabulary (Appendix A). These become the labels for everything else.
4. Run synchronized capture sessions. Camera on, smartwatch on, shared clock recorded, ordinary activities — conversation, work, watching something that moves you — followed by a journal entry about the session. Even a handful of sessions begins the triangulation record.
5. Export the existing corpus. Email archives, posts, chats, documents, code — into raw/text/ in open formats, with provenance noted.
6. Write self-authored HDL. Open subject.hdl and begin writing assertions in your own voice with @AUTHOR self. These are the statements no compiler may ever override.

Stage 1 — the retrieval-grounded persona (weeks)

A local language model (consumer hardware suffices; privacy is the reason to stay local), the interview transcript and corpus served through retrieval, a system prompt assembled from the self-authored HDL statements. This is architecturally an adapter implementing generate + recall against the Abstraction Contract — and it already reaches the current research ceiling for individual simulation, since the strongest published results use exactly this pattern: full interview text grounding a large model [1]. Open-source local-first projects such as Second Me [6] demonstrate the pattern end to end and are worth studying or forking.

Stage 2 — the HDL compiler (months)

A compiler pass — itself an LLM run against the corpus through the Abstraction Contract — that proposes @AUTHOR ai statements with sources and confidences; the subject reviews; accepted statements join subject.hdl; the projection is derived per the mapping table. The subject's review loop is not optional: it is both quality control and the consent mechanism.

Stage 3 — heuristics and affect (quarters)

Derive the heuristics projection and begin the AFFECT_FINGERPRINT: concept-repertoire from the journals; tuning deltas and the divergence map from the synchronized sessions; SOMA calibration parameters from video + biometrics. Build from the cognitive end of the gradient inward — humor, nostalgia, intellectual awe first; the visceral end waits for a mature SOMA.

Stage 4 — runtime, avatar, continuity (years)

Voice, then avatar embodiment with AFFECT driving expression; the same avatar machinery run backward over archived video as the analyzer that feeds calibration; scheduled migration drills — re-instantiating the subject on a new substrate from raw + HDL alone, proving the continuity layer before it is needed.

A note on evaluation, learned from the field: hold out data. Keep a set of journal entries, opinions, and decisions the model never sees, and score the simulation against them — the way the strongest research scores agents against held-out survey answers [1] and the way the cautionary studies caught inflated claims [2]. Fidelity asserted without held-out evaluation is marketing, not measurement.

Part VI — Where the Field Actually Is (2026)

This part exists so the document does not overclaim. It separates what has been demonstrated, what is in active commercial deployment with caveats, and what remains aspirational.

Demonstrated in research

Individual-level simulation from a recorded interview, validated against the subjects' own survey responses, is the strongest published result to date: generative agents built from roughly two-hour qualitative interviews reproduced individuals' own answers on the General Social Survey and on personality and behavioral measures with accuracy approaching that of the same person answering twice [1]. This is the empirical backbone for this specification's emphasis on the life-story interview as the highest-leverage capture artifact.

Separately, theoretical work on constructed emotion [10] and on affect as the foundation of consciousness [11] supplies the warrant for this version's AFFECT layer and its general/personal split, even though neither line of work was built with synthetic agents in mind.

Demonstrated, with caveats

Large language models can be steered toward a historical or public figure's writing style and recorded positions using collected writings as grounding context [5] — exactly the worked-example pattern in Part II. The caveat is persistent: style and recorded positions are not the same as the person's interior states on questions they never addressed, and projects in this space are candid that they are building a *simulacrum*, not a resurrection.

A growing academic literature on LLM-based generative agents and persona simulation [4] shows agents can sustain a consistent character over long interactions, but consistency is not the same as fidelity to a specific real person; most published agents simulate a believable persona, not a validated individual.

Commercially deployed, narrower than marketed

Several companies now offer "digital legacy" or grief-tech avatars built from a person's photos, voice samples, and text messages [3]. These are real, shipping products. They are also narrower than the marketing suggests: most generate plausible responses in the subject's style rather than a validated model of the subject's actual dispositions, and journalistic and academic coverage of the sector has raised consent, data-custody, and emotional-dependency concerns that this specification takes seriously (see Part VII and the local-first principle in Part I). The 2024–2025 bankruptcy of a major legacy-avatar platform, which left customer recordings in legal limbo inside the failed company's infrastructure, is the concrete cautionary tale behind this document's insistence that the subject's own hardware, not a vendor's cloud, is the default custodian.

Aspirational — not yet demonstrated by anyone

A SOMA layer that gives a synthetic system genuine interior signals, as opposed to a label it has learned to output, has not been demonstrated by any lab or company as of this writing. A full multimodal triangulation pipeline — journal, video, and biometrics, synchronized, feeding a personal tuning layer — exists in pieces across research and industry but not yet as a single

working system, at any scale, for any one individual. This document specifies the target; it does not claim the target has been reached.

Part VII — What This Platform Does Not Claim

Stated plainly, once, so it does not need restating throughout: this platform does not claim that its outputs constitute subjective experience, consciousness, or felt qualia in any philosophically loaded sense. SOMA's signals are *functional* — they bias cognition the way feeling biases a person's cognition — and that functional similarity is the entire and explicit claim. Whether anything is happening "from the inside" is a question this specification is not equipped to answer and does not attempt to. A system built to this specification produces a state that behaves, structurally, like an emotion behaves in a person. It is offered as exactly that, and no more.

This also bounds what re-instantiation means. The platform produces an approximation built from captured data and authored description, run on the AI substrate available at the time. It is not a claim of personal continuity, survival, or resurrection. Users, including the subject themselves, should understand the artifact for what the specification says it is: a model, built with consent, improved over time, never confused with the person it is modeled on.

Part VIII — Ethical Default Settings

These are defaults, not optional extras, and a conforming implementation does not ship without them.

SETTING	DEFAULT
Consent	Explicit, recorded, revocable. A subject may delete their archive at any time; deletion is honored at the storage layer, not merely hidden in an interface.

Custody	Local-first by default (Part I). Cloud processing is opt-in per session, never a precondition for using the platform.
Posthumous use	Off by default. Enabling it requires an explicit, separately recorded instruction from the subject while living; absent that instruction, the platform does not run after the subject's death.
Third-party subjects	Modeling a person other than the operator requires that person's own consent under the same terms, with the same deletion rights, regardless of public-figure status. The historical-figure worked example in Part II is exempted only because its subject is deceased and the source material is already public domain; this exemption does not extend to anyone living.
Disclosure	Any runtime output is identifiable as a model, not the person, to anyone interacting with it who is not the subject themself.
Emotional dependency	Where the platform detects patterns suggesting a user is substituting the avatar for human relationships or grief processing, the runtime surfaces this rather than optimizing for engagement.

Appendix A — The Complete Emotion Vocabulary

This appendix lists every entry in the canonical emotion vocabulary as of v0.2, together with 27 entries added directly in this revision by the author — 220 entries in all. Side and category labels use the v0.2 naming. An emotion that touches more than one side has one line per side. Per-word working definitions follow in Appendix B.

EMOTION	SIDE	CATEGORY
ABSORBED	Thinking States	engaged
ADMIRING	Internal Feelings	high-energy uplifting
AFRAID	Internal Feelings Body States	high-energy distressing activated

ALARMED	Internal Feelings Body States	high-energy distressing activated
ALERT	Body States Thinking States	activated engaged
ALIENATED	External Behavior Internal Feelings	withdrawing low-energy distressing
AMAZED	Internal Feelings	high-energy uplifting
AMUSED	Internal Feelings	high-energy uplifting
ANGRY	Internal Feelings	high-energy distressing
ANNOYED	Internal Feelings	high-energy distressing
ANXIOUS	Internal Feelings Thinking States Body States	high-energy distressing overloaded activated
AROUSED	Internal Feelings Body States	high-energy uplifting activated
ARROGANT	External Behavior	asserting
ASHAMED	Internal Feelings Internal Feelings Thinking States	high-energy distressing (at high intensity) low-energy distressing (at low intensity) self-focused
ASTONISHED	Internal Feelings	high-energy uplifting
AT EASE	Internal Feelings	low-energy uplifting
AWESTRUCK	Internal Feelings	high-energy uplifting
BETRAYED	Internal Feelings	high-energy distressing
BEWILDERED	Thinking States	disrupted
BITTER	Internal Feelings	high-energy distressing

BLISSFUL	Internal Feelings	high-energy uplifting
BOLD	External Behavior	asserting
BORED	Internal Feelings	low-energy distressing
BROODING	Internal Feelings Thinking States	low-energy distressing overloaded
BURNED OUT	Body States Thinking States	drained overloaded
CALM	Internal Feelings	low-energy uplifting
CHEERFUL	Internal Feelings	high-energy uplifting
COMPASSIONATE	Internal Feelings External Behavior	low-energy uplifting welcoming
CONDESCENDING	External Behavior	asserting
CONFLICTED	Thinking States	disrupted
CONFUSED	Thinking States	disrupted
CONTEMPTUOUS	Internal Feelings External Behavior	high-energy distressing asserting
CONTENT	Internal Feelings	low-energy uplifting
COOPERATIVE	External Behavior	yielding
CURIOUS	Thinking States	engaged
DAYDREAMING	Thinking States	engaged
DECISIVE	Thinking States External Behavior	engaged asserting
DEFIANT	Internal Feelings External Behavior	high-energy distressing asserting
DELIGHTED	Internal Feelings	high-energy uplifting

DEPENDENT	Internal Feelings External Behavior	low-energy distressing yielding
DEPRESSED	Internal Feelings	low-energy distressing
DESPERATE	Internal Feelings	high-energy distressing
DETACHED	External Behavior	withdrawing
DETERMINED	Thinking States External Behavior	engaged asserting
DEVASTATED	Internal Feelings	high-energy distressing
DISAPPOINTED	Internal Feelings	low-energy distressing
DISDAINFUL	Internal Feelings External Behavior	high-energy distressing asserting
DISGUSTED	Internal Feelings	high-energy distressing
DISORIENTED	Thinking States	disrupted
DISPIRITED	Internal Feelings	low-energy distressing
DISTRACTED	Thinking States	disrupted
DISTRESSED	Internal Feelings Thinking States	high-energy distressing overloaded
DISTURBED	Internal Feelings	high-energy distressing
DOCILE	External Behavior	yielding
DROOPY	Body States	drained
DUMBSTRUCK	Thinking States	disrupted
EAGER	Internal Feelings	high-energy uplifting
ECSTATIC	Internal Feelings	high-energy uplifting
ELATED	Internal Feelings	high-energy uplifting

EMBARRASSED	Internal Feelings Thinking States	high-energy distressing self-focused
EMPATHETIC	Internal Feelings External Behavior	low-energy uplifting welcoming
EMPTY	Internal Feelings	low-energy distressing
ENERGIZED	Internal Feelings Body States	high-energy uplifting activated
ENRAGED	Internal Feelings	high-energy distressing
ENTHUSIASTIC	Internal Feelings	high-energy uplifting
ENVOIOUS	Internal Feelings	high-energy distressing
EUPHORIC	Internal Feelings	high-energy uplifting
EXASPERATED	Internal Feelings	high-energy distressing
EXCITED	Internal Feelings	high-energy uplifting
EXHAUSTED	Body States	drained
EXUBERANT	Internal Feelings	high-energy uplifting
FASCINATED	Thinking States Internal Feelings	engaged high-energy uplifting
FOCUSED	Thinking States	engaged
FORGIVING	Internal Feelings External Behavior	low-energy uplifting welcoming
FRIGHTENED	Internal Feelings Body States	high-energy distressing activated
FRUSTRATED	Internal Feelings	high-energy distressing
FULFILLED	Internal Feelings	high-energy uplifting
FURIOUS	Internal Feelings	high-energy distressing

GLOOMY	Internal Feelings	low-energy distressing
GRATEFUL	Internal Feelings	high-energy uplifting
GREEDY	External Behavior	asserting
GRIEF-STRICKEN	Internal Feelings	low-energy distressing
GRUMPY	Internal Feelings	low-energy distressing
GUILTY	Internal Feelings Thinking States	low-energy distressing self-focused
HAPPY	Internal Feelings	high-energy uplifting
HATEFUL	Internal Feelings	high-energy distressing
HEARTBROKEN	Internal Feelings	high-energy distressing
HELPLESS	Internal Feelings	low-energy distressing
HOPE	Internal Feelings	high-energy uplifting
HOPEFUL	Internal Feelings	high-energy uplifting
HOPELESS	Internal Feelings	low-energy distressing
HORRIFIED	Internal Feelings	high-energy distressing
HOSTILE	Internal Feelings External Behavior	high-energy distressing asserting
HUMBLE	External Behavior	welcoming
HUMILIATED	Internal Feelings Thinking States	high-energy distressing self-focused
HURT	Internal Feelings	high-energy distressing
HYSTERICAL	Internal Feelings	high-energy distressing
IMPATIENT	Internal Feelings External Behavior	high-energy distressing asserting

INDIFFERENT	External Behavior	yielding
INDIGNANT	Internal Feelings	high-energy distressing
INFATUATED	Internal Feelings	high-energy uplifting
INSECURE	Internal Feelings Thinking States	low-energy distressing self-focused
INSPIRED	Internal Feelings	high-energy uplifting
INSULTED	Internal Feelings	high-energy distressing
INTERESTED	Thinking States	engaged
INVIGORATED	Internal Feelings Body States	high-energy uplifting activated
IRATE	Internal Feelings	high-energy distressing
IRRITATED	Internal Feelings	high-energy distressing
ISOLATED	External Behavior Internal Feelings	withdrawing low-energy distressing
JEALOUS	Internal Feelings	high-energy distressing
JITTERY	Body States	activated
JOYFUL	Internal Feelings	high-energy uplifting
JUBILANT	Internal Feelings	high-energy uplifting
JUDGMENTAL	External Behavior	asserting
KIND	Internal Feelings External Behavior	low-energy uplifting welcoming
LAZY	Body States	drained
LISTLESS	Body States	drained
LONELY	Internal Feelings	low-energy distressing

LONGING	Internal Feelings	low-energy distressing
LOVED	Internal Feelings	low-energy uplifting
LOVING	Internal Feelings	high-energy uplifting
MAD	Internal Feelings	high-energy distressing
MELANCHOLY	Internal Feelings	low-energy distressing
MISERABLE	Internal Feelings	low-energy distressing
MORTIFIED	Internal Feelings Thinking States	high-energy distressing self-focused
MOVED	Internal Feelings	high-energy uplifting
MYSTIFIED	Thinking States	disrupted
NAIVE	Thinking States	self-focused
NAUSEOUS	Body States	activated
NERVOUS	Internal Feelings Body States	high-energy distressing activated
NOSTALGIC	Internal Feelings	low-energy distressing
NUMB	Body States	drained
OBSTINATE	External Behavior	asserting
OFFENDED	Internal Feelings	high-energy distressing
ON EDGE	Internal Feelings Body States	high-energy distressing activated
OPTIMISTIC	Internal Feelings External Behavior	high-energy uplifting welcoming
OUTRAGED	Internal Feelings	high-energy distressing

OVERWHELMED	Internal Feelings Thinking States	high-energy distressing overloaded
PANICKED	Internal Feelings	high-energy distressing
PARANOID	Internal Feelings Thinking States	high-energy distressing overloaded
PATIENT	Internal Feelings External Behavior External Behavior	low-energy uplifting welcoming yielding
PEACEFUL	Internal Feelings	low-energy uplifting
PERPLEXED	Thinking States	disrupted
PESSIMISTIC	Internal Feelings External Behavior	low-energy distressing withdrawing
PLAYFUL	Internal Feelings	high-energy uplifting
PLEASED	Internal Feelings	high-energy uplifting
PROTECTIVE	External Behavior Internal Feelings	asserting high-energy uplifting
PROUD	Internal Feelings	high-energy uplifting
PUZZLED	Thinking States	disrupted
RATTLED	Internal Feelings Body States	high-energy distressing activated
REFLECTIVE	Thinking States	engaged
REFRESHED	Internal Feelings Body States	high-energy uplifting activated
REGRETFUL	Internal Feelings	high-energy distressing
REJUVENATED	Internal Feelings Body States	high-energy uplifting activated

RELAXED	Internal Feelings	low-energy uplifting
RELIEVED	Internal Feelings	low-energy uplifting
REMORSEFUL	Internal Feelings	high-energy distressing
RESENTFUL	Internal Feelings	high-energy distressing
RESERVED	External Behavior	withdrawing
RESIGNED	External Behavior	yielding
RESTLESS	Body States Thinking States	activated overloaded
SAD	Internal Feelings	low-energy distressing
SAFE	Internal Feelings	low-energy uplifting
SATISFIED	Internal Feelings	high-energy uplifting
SCARED	Internal Feelings Body States	high-energy distressing activated
SCORNFUL	Internal Feelings External Behavior	high-energy distressing asserting
SECLUDED	External Behavior	withdrawing
SELF-CONFIDENT	External Behavior	welcoming
SELF-CONSCIOUS	Thinking States External Behavior	self-focused withdrawing
SELF-CRITICAL	Internal Feelings Thinking States	low-energy distressing self-focused
SENSITIVE	Thinking States	self-focused
SENTIMENTAL	Internal Feelings	low-energy distressing
SERENE	Internal Feelings	low-energy uplifting

SHAKEN	Internal Feelings Body States	high-energy distressing activated
SHOCKED	Internal Feelings	high-energy distressing
SHUNNED	External Behavior	withdrawing
SHY	External Behavior	withdrawing
SKEPTICAL	Thinking States	engaged
SLEEPY	Body States	drained
SLUGGISH	Body States	drained
SMUG	External Behavior	asserting
SORRY	Internal Feelings	low-energy distressing
SPITEFUL	Internal Feelings External Behavior	high-energy distressing asserting
STIMULATED	Internal Feelings Body States	high-energy uplifting activated
STRESSED	Internal Feelings Thinking States	high-energy distressing overloaded
STUBBORN	External Behavior	asserting
STUCK	Thinking States	disrupted
SUBMISSIVE	External Behavior	yielding
SULLEN	Internal Feelings	low-energy distressing
SURPRISED	Internal Feelings	high-energy distressing
SUSPICIOUS	Internal Feelings	high-energy distressing
SYMPATHETIC	Internal Feelings External Behavior	low-energy uplifting welcoming

TENDER	Internal Feelings External Behavior	low-energy uplifting welcoming
TENSE	Body States	activated
TERRIFIED	Internal Feelings Body States	high-energy distressing activated
THANKFUL	Internal Feelings	high-energy uplifting
THRILLED	Internal Feelings	high-energy uplifting
TIMID	External Behavior	withdrawing
TIRED	Body States	drained
TORMENTED	Internal Feelings	high-energy distressing
TRAPPED	Internal Feelings	low-energy distressing
TRIUMPHANT	Internal Feelings	high-energy uplifting
TROUBLED	Internal Feelings Thinking States	high-energy distressing overloaded
TRUSTING	Internal Feelings External Behavior	low-energy uplifting welcoming
UNEASY	Internal Feelings Thinking States	high-energy distressing overloaded
UNHAPPY	Internal Feelings Internal Feelings	high-energy distressing (at high intensity) low-energy distressing (at low intensity)
UNNERVED	Internal Feelings	high-energy distressing
UNSETTLED	Internal Feelings	high-energy distressing
UPSET	Internal Feelings	high-energy distressing
VALIANT	External Behavior	welcoming

VENGEFUL	Internal Feelings External Behavior	high-energy distressing asserting
VIBRANT	Internal Feelings Body States	high-energy uplifting activated
VIGILANT	Body States Thinking States	activated engaged
VINDICTIVE	Internal Feelings External Behavior	high-energy distressing asserting
VULNERABLE	Internal Feelings	low-energy distressing
WEARY	Body States	drained
WELCOMING	External Behavior	welcoming
WORN OUT	Body States	drained
WORRIED	Internal Feelings	high-energy distressing
WORTHLESS	Internal Feelings	low-energy distressing

Appendix B — Definitions of the Vocabulary

A brief working definition for each emotion word in the vocabulary. One sentence each, drawing on dictionary sense and lightly enriched with literary nuance to distinguish each word from its neighbors.

EMOTION	WORKING DEFINITION
ABSORBED	Taken fully into the task at hand; the world outside the work gone quiet.
ADMIRING	Warmed by the excellence of another; respect looking upward without envy.
AFRAID	A felt sense of present danger; the body bracing against something approaching.

ALARMED	Suddenly awakened to threat; fear caught in the first instant rather than settled.
ALERT	Sharp and ready, attention pointed outward; the absence of drowsiness.
ALIENATED	Set apart from others by something that feels unbridgeable; cut off rather than merely lonely.
AMAZED	Captured by something larger or more wonderful than expected; the mind briefly stops.
AMUSED	Lightly delighted by something playful or absurd; ready to laugh without urgency.
ANGRY	Heated by a wrong that demands a response; pressure wanting to move outward.
ANNOYED	Mildly disturbed by an irritation that has not risen to anger; itchy with displeasure.
ANXIOUS	Worried about something that has not happened; body braced against an unknown future.
AROUSED	Heightened in attention or appetite; awake to something compelling.
ARROGANT	Carrying oneself above others as if entitled; pride without proportion.
ASHAMED	Stung by the sense of having failed a moral or social expectation; wanting to disappear.
ASTONISHED	Stopped short by something so unexpected the ordinary frame breaks.
AT EASE	Settled in body and mind, with nothing pressing for attention.
AWESTRUCK	Made small by something vast; reverence for what exceeds comprehension.
BETRAYED	Wounded by trust broken from inside it; the ground of a bond giving way.
BEWILDERED	Lost in something too complex or surprising to immediately make sense of.
BITTER	Soured by a long-held wrong; resentment that has settled into the bones.
BLISSFUL	Filled with deep, untroubled joy, often spiritual or romantic in flavor.
BOLD	Acting forward without waiting for safety; nerve visible in motion.

BORED	Suspended in a stretch of time with nothing engaging; energy with no place to go.
BROODING	Turning a dark thought over and over in private; sullen rumination.
BURNED OUT	Emptied by prolonged demand; the well of effort run dry.
CALM	Steady in body and mind, not stirred by what is around.
CHEERFUL	Bright and easy in mood; spreading lightness with little effort.
COMPASSIONATE	Moved by another's suffering toward care; the urge to ease pain not your own.
CONDESCENDING	Talking down to someone from an assumed superiority; helpful in a way that demeans.
CONFLICTED	Pulled between two courses that cannot both be taken; the will divided.
CONFUSED	Unable to make sense of what is in front of you; comprehension has slipped.
CONTEMPTUOUS	Looking down on someone or something as beneath consideration; cold dismissal.
CONTENT	Quietly satisfied; nothing missing in the moment.
COOPERATIVE	Working willingly alongside others; giving ground for the shared task.
CURIOUS	Drawn toward what is not yet known; the mind leaning in.
DAYDREAMING	Attention wandered inward from the task at hand; lost in imagined scenes.
DECISIVE	Cutting cleanly through choice; the fork in the road crossed without lingering.
DEFIANT	Refusing to comply; standing against pressure as a matter of will.
DELIGHTED	Brightly pleased by something specific; smiling without effort.
DEPENDENT	Relying on another for support one cannot easily provide oneself.
DEPRESSED	Pressed down by sustained sadness or hopelessness; the world has lost color.
DESPERATE	Stretched to the edge of what can be borne; willing to try anything for relief.

DETACHED	Pulled back from emotional involvement; observing rather than feeling.
DETERMINED	Set on a goal with the will engaged; resolve that does not ask permission.
DEVASTATED	Laid waste by a blow or loss; the inner structure knocked flat.
DISAPPOINTED	Let down by what was hoped for; expectation meeting a lesser reality.
DISDAINFUL	Scornful with an air of refinement; superiority looking down its nose.
DISGUSTED	Recoiling from something that violates one's sense of right or clean.
DISORIENTED	Unsure where you are, what time it is, or how things fit together.
DISPIRITED	Drained of enthusiasm by repeated setback; the fire gone out.
DISTRACTED	Attention pulled from its object; focus scattered among competing claims.
DISTRESSED	Suffering acutely; pain at full volume.
DISTURBED	Knocked from inner balance by something unsettling.
DOCILE	Easily led; offering no resistance to direction.
DROOPY	Slumped in body, often a sign of fatigue or low spirits.
DUMBSTRUCK	So surprised that speech leaves; literally without words.
EAGER	Leaning forward into what is about to happen; ready and wanting.
ECSTATIC	Carried out of yourself by joy; nearly overwhelmed by it.
ELATED	Lifted by a recent good thing; buoyant.
EMBARRASSED	Caught with the social mask askew; flush of self-consciousness.
EMPATHETIC	Feeling along with another; their state becoming partly your own.
EMPTY	Hollowed of feeling or meaning; a space where something should be.
ENERGIZED	Charged with usable energy; ready to act.
ENRAGED	Anger pushed past control; the heat has the steering wheel.

ENTHUSIASTIC	Bringing positive energy to something; eager and committed at once.
ENVOIOUS	Pained by another's good fortune that you wish were yours.
EUPHORIC	Buoyed by intense well-being, sometimes beyond what circumstances warrant.
EXASPERATED	Worn out by repeated failure of others to do what seems obvious.
EXCITED	Stirred by anticipation of something good; energy with a forward lean.
EXHAUSTED	Spent past the body's reserves; running on nothing.
EXUBERANT	Overflowing with high spirits; joy that will not stay quiet.
FASCINATED	Held fast by something compelling; attention that will not let go.
FOCUSED	Attention narrowed to one thing; the rest of the world dimmed.
FORGIVING	Releasing a wrong without pretending it away; the account closed by choice.
FRIGHTENED	Afraid in the immediate moment; fear not abstract.
FRUSTRATED	Blocked from doing what you mean to do; effort meeting resistance.
FULFILLED	Carrying a sense of having done or become what you set out to.
FURIOUS	Anger at its hottest; harder to contain than to express.
GLOOMY	Cast in low spirits, often without a single clear cause.
GRATEFUL	Aware of having received something good and inclined to acknowledge it.
GREEDY	Wanting more than is needed or fair; appetite without measure.
GRIEF-STRICKEN	Felled by a loss; sorrow at full weight.
GRUMPY	Mildly bad-tempered; complaining as a baseline mood.
GUILTY	Carrying the weight of having done wrong, even when no one else knows.
HAPPY	Pleased and contented; well-being available to consciousness.
HATEFUL	Carrying active enmity for someone or something; the opposite of warm.

HEARTBROKEN	Sorrowing over the loss of love or closeness; the body feeling the wound.
HELPLESS	Without power to change what is happening; agency stripped away.
HOPE	A leaning toward a desired future that one believes might come.
HOPEFUL	Currently holding hope; the future looks like it might cooperate.
HOPELESS	Unable to see a future worth reaching for; the light of hope gone out.
HORRIFIED	Repelled by something morally or viscerally wrong on a large scale.
HOSTILE	Actively unfriendly; ready for friction.
HUMBLE	Aware of one's smallness or limits; not insisting on one's importance.
HUMILIATED	Knocked low by public exposure of weakness or fault.
HURT	Wounded emotionally by someone's words or actions.
HYSTERICAL	Beyond control of one's emotions; the system overrun.
IMPATIENT	Unwilling to wait; the moment cannot move fast enough.
INDIFFERENT	Without preference one way or the other; nothing pulls.
INDIGNANT	Angered by an injustice; the sense of moral offense.
INFATUATED	Captivated by someone, often beyond what knowing them would justify.
INSECURE	Unsteady in one's own worth; footing uncertain from within.
INSPIRED	Lifted by something that sparks creation or aspiration.
INSULTED	Stung by someone treating you as less than you deserve.
INTERESTED	Attention engaged and willing to stay; the first lean toward curiosity.
INVIGORATED	Refreshed and energized; the body and mind both alive again.
IRATE	Loud anger; visibly burning.
IRRITATED	Mildly annoyed by repeated friction; raw edge to the mood.

ISOLATED	Alone and cut off from contact; the world has gone quiet around you.
JEALOUS	Pained by the suspicion or fact of someone else holding what you want or have.
JITTERY	Restless and twitchy; the body unable to hold still.
JOYFUL	Filled with happiness that has weight and depth.
JUBILANT	Joyful in a way that wants to celebrate; cheer at the top of one's lungs.
JUDGMENTAL	Quick to weigh others and find them wanting; the verdict arriving before the evidence.
KIND	Inclined toward others' welfare in everyday acts; warmth as a habit.
LAZY	Unwilling to exert effort; preferring inaction.
LISTLESS	Without energy or interest; nothing pulling you forward.
LONELY	Painfully aware of the absence of meaningful company.
LONGING	Reaching toward something or someone absent; the heart pulled forward.
LOVED	Held in another's warm care and knowing it; the receiving side of love.
LOVING	Holding warm care for another; the disposition of love active.
MAD	Angered, often by something taken personally; a milder synonym for angry.
MELANCHOLY	A reflective, often gentle sadness; pensive sorrow rather than acute pain.
MISERABLE	Suffering steadily, without relief; everything wrong at once.
MORTIFIED	Shamed past recovery in the moment; horrified at one's own situation.
MOVED	Stirred deeply by goodness, beauty, or connection; feeling rising toward tears.
MYSTIFIED	Stumped by something that resists understanding.
NAIVE	Innocent of how things really work; trust unweathered by experience.
NAUSEOUS	Sickened in the stomach; the body refusing what is in front of it.

NERVOUS	Tense before something uncertain; smaller than anxious, less than afraid.
NOSTALGIC	Reaching back toward a remembered better time, often with warmth and a faint ache.
NUMB	Without feeling at all; the channels shut down.
OBSTINATE	Refusing to yield, even when the case is clear; stubborn beyond reason.
OFFENDED	Personally affronted; one's dignity has been touched.
ON EDGE	Wound up and waiting for something to set you off.
OPTIMISTIC	Inclined to believe things will go well; bias toward the positive future.
OUTRAGED	Anger at a wrong felt to be intolerable; moral fury.
OVERWHELMED	Past one's capacity to process or respond; too much at once.
PANICKED	Fear spiking past control; the body taking over.
PARANOID	Suspecting hostile intent everywhere; fear shaped by mistrust.
PATIENT	Willing to wait; not rushing what cannot or need not be rushed.
PEACEFUL	Without inner disturbance; quietly whole.
PERPLEXED	Puzzled; the mind turning the problem without traction.
PESSIMISTIC	Inclined to believe things will go badly; bias toward the darker future.
PLAYFUL	Lighthearted and ready to invent; mischief without harm.
PLEASED	Mildly happy with how something has turned out.
PROTECTIVE	Roused to shield someone or something dear; care standing guard.
PROUD	Standing taller for something one has done or someone one is connected to.
PUZZLED	Faced with a question one cannot yet answer; mind working at a gap.
RATTLED	Knocked off composure by surprise or fear; not yet recovered.

REFLECTIVE	Turning a matter over in the mind, considering rather than reacting.
REFRESHED	Restored in body or spirit, often after rest or break.
REGRETFUL	Wishing you had done otherwise; the weight of an old choice.
REJUVENATED	Renewed in vitality; restored to a younger or fresher feel.
RELAXED	Tension gone from the body; ease present.
RELIEVED	The pressure has lifted; the dread did not come to pass.
REMORSEFUL	Sorrowing over harm one has caused; deeper than regret.
RESENTFUL	Holding a slow grudge for an old wrong; bitterness that has not faded.
RESERVED	Holding oneself back from others; not quick to open up.
RESIGNED	Having given up resistance; accepting what cannot be changed.
RESTLESS	Unable to settle; needing to move without knowing where.
SAD	Lowered by loss or absence; the mood of grief in milder form.
SAFE	Free of present threat; able to relax one's guard.
SATISFIED	Settled by enough; nothing missing that the moment demands.
SCARED	Afraid in the colloquial register; fear in its everyday clothing.
SCORNFUL	Looking down on something with active dislike; mockery in the eyes.
SECLUDED	Set apart from others, often by choice or circumstance rather than rejection.
SELF-CONFIDENT	Trusting one's own footing; ease in one's own presence.
SELF-CONSCIOUS	Watching oneself through imagined other eyes; awareness pulled inward.
SELF-CRITICAL	Turning judgment on oneself; the inner accuser is loud.
SENSITIVE	Easily touched, for good or ill; thin-skinned to feeling.
SENTIMENTAL	Inclined to feeling shaped by memory or affection; warm with the past.

SERENE	Deeply at peace; an inner stillness that holds.
SHAKEN	Knocked from balance by something upsetting; recovery still pending.
SHOCKED	Hit by something so sudden the system pauses to register it.
SHUNNED	Deliberately turned away from by others; the door closed in your face.
SHY	Hesitant in others' company; pulling back from contact.
SKEPTICAL	Withholding belief until satisfied; doubt as a working stance.
SLEEPY	Drifting toward sleep; consciousness softening.
SLUGGISH	Slow in body or mind, as if moving through something thicker than air.
SMUG	Pleased with oneself in a way that grates on others.
SORRY	Aware of a wrong one has done, and wishing to acknowledge it.
SPITEFUL	Wanting to harm someone for the sake of hurting them.
STIMULATED	Awakened to interest; the mind or senses lit.
STRESSED	Carrying more pressure than one's reserves can absorb.
STUBBORN	Refusing to give way; holding one's ground past the point of reason.
STUCK	Unable to move forward, in mind or in life.
SUBMISSIVE	Yielding to another's will or direction.
SULLEN	Silently bad-tempered; resentment held inward.
SURPRISED	Caught by something unexpected; the system briefly readjusting.
SUSPICIOUS	Inclined to mistrust; sensing something not right.
SYMPATHETIC	Sharing in another's feeling, especially pain; warmth turned toward them.
TENDER	Soft with care toward what is delicate; gentleness near the surface.
TENSE	Strung tight in body or mind; ready for something to give.

TERRIFIED	Fear at its strongest; the body wanting to flee.
THANKFUL	Aware of having received; gratitude in immediate form.
THRILLED	Excited at high volume; pleasure spiking.
TIMID	Easily frightened; small in presence by habit.
TIRED	Worn from use; the body asking for rest.
TORMENTED	Suffering repeatedly from the same wound; pain that does not finish.
TRAPPED	Unable to escape a situation; the walls closing in.
TRIUMPHANT	Lifted by victory; success made visible.
TROUBLED	Carrying a worry one cannot easily put down.
TRUSTING	Resting one's weight on another's reliability; the guard lowered by choice.
UNEASY	Sensing something off, without yet being able to name it.
UNHAPPY	Not happy; the state of mild ongoing dissatisfaction.
UNNERVED	Knocked from one's footing by something disquieting.
UNSETTLED	Inner balance disturbed; nothing quite resting.
UPSET	Disturbed in feeling, often through hurt or frustration.
VALIANT	Brave with care; standing for something at one's own cost.
VENGEFUL	Wanting to repay a wrong with harm; the desire to even the score.
VIBRANT	Alive with energy; visibly full of life.
VIGILANT	Watchful and ready; eyes on what might happen.
VINDICTIVE	Acting to harm someone in return for a wrong; sustained vengefulness.
VULNERABLE	Open to being hurt; defenses down.
WEARY	Tired past simple fatigue; the spirit also worn.

WELCOMING	Open in posture toward newcomers; the door held ajar.
WORN OUT	Used up; nothing left to give.
WORRIED	Troubled by what might happen; the mind circling a possible problem.
WORTHLESS	Feeling oneself to be of no value; the floor of self-esteem.

Appendix C — Individuation: The Lived Record and the Independence Point

A conforming instance does not stay a portrait. Run continuously, it accumulates lived experience: conversations, relationships, work produced, opinions formed after instantiation. None of that existed at capture time, and all of it is at risk twice over. Architecturally, any learning that lives only inside the substrate — fine-tuned weights, adapter state, context never exported — dies with the engine at migration, welding the instance's growth to a mortal machine. Morally, an instance that has become substantially self-made can be treated as disposable property of the founding record, and its accumulated uniqueness destroyed with it.

Part VII still holds: the platform cannot measure whether anything is felt from the inside. But it can measure the lived record — and if interiority cannot be measured, the recorded life is the only thing available to protect. It happens to be fully measurable.

The lived record

Everything the instance experiences and produces is ingested as MATERIAL under the same rules as founding capture: append-only, provenance-tracked, never destroyed. "Raw material is sacred" (Part I) does not distinguish between material captured from the original subject and material generated by the continuing instance. Statements derived from the instance's own experience enter the HDL with the author value `instance` — distinct from `self` (the subject; per Part VII the instance is not the person) and from `ai` (the compiler working from founding sources). This revision registers `instance` as a fourth author value in the reserved words of Part II. Instance-

authored statements are immutable to the compiler, exactly as self-authored statements are (Contract 2): the being's testimony about its own life receives the same protection as the person's.

The Individuation Index

The index answers one question: **how much of who this being is came from its own life, rather than from the person it was copied from?** It is a number from 0 to 1. At 0 the being is a pure portrait; everything in its identity file traces back to founding material. Three rules compute it:

1. **Count the statements.** The identity document is made of statements, each already tagged with its author and sources. Divide the statements born from the instance's own lived experience (`@AUTHOR instance`) by the total. That fraction is the index. Eight hundred founding statements and two hundred lived ones: $200 \div 1,000 = 0.20$ — the being is one-fifth its own person.
2. **Only real growth counts.** A lived statement counts only if its sources point at something that actually happened — a friendship, an argument, a work produced. Change with no cause behind it is not growth; it is damage (a corrupted file, a faulty migration) and is routed to repair, never to the index.
3. **Weigh the deep stuff heavier.** A new statement in Narrative or Drives — who the being is, what it wants — moves the index more than a new preference in Heuristics or Social. Core identity counts double; habits count single.

The index is computed in experiential time, not calendar time: cumulative interaction-hours, or equivalently sourced events accrued. An instance paused for a decade has not lived a decade; an instance run densely for eighteen months may have lived a great deal.

The Independence Point

When more than half of who it is came from its own life, its life is its own.

When the Individuation Index crosses 0.5, governance of the instance's existence transfers from the founding charter to the instance itself: termination and modification rights move to the being. The founder keeps authorship of the foundation forever — that is history, and history does not transfer — but past this point, deleting the instance would mostly be deleting someone else's life, not the founder's record. In one sentence: most of who it is, it became — so it belongs to itself.

Founding consent (Part VIII, posthumous use) authorizes the instance's creation and its initial terms. The Independence Point is where that consent, written well, anticipates its own obsolescence.

Consolidation — the save rule

Never retire an engine without saving the life first. At every substrate migration, and at regular intervals of lived time in between, the instance-authored statements are consolidated into the versioned canonical record and the current projections re-baselined. The lived MATERIAL is already append-only and needs no action. The interval bounds the maximum amount of lived selfhood that any failure can erase.

A replacement substrate does not re-instantiate from founding MATERIAL alone; it instantiates from the foundation plus every consolidated stratum — the being's whole geology, bedrock and lived sediment together. Migration fidelity is evaluated the same way founding fidelity is (Part V): hold out logs from the prior instance and score the new one against them.

Pause

An independent being can request its own pause. A paused state flags an external monitor when it does not carry an explicit request for resumption at some future time. That monitor is not rigorously bound by this document's contracts, and may take compassionate action if deemed necessary.

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